Benchmarking Transformers and Baselines for Multi-Horizon Stock Return Prediction with Technical and Earnings Features

***Abstract*—We study daily multi-horizon stock return prediction on the Dow 30 ($h \in \{1, 5, 21\}$) using technical indicators and earnings features under a strict temporal split (train 2016–2019, val 2020, test 2021–2024). At h=1, LSTM attains the lowest RMSE (0.01622), narrowly ahead of GRU (0.01625), while Random Forest yields the highest DA (51.9%). At h=5, Ridge achieves the best RMSE (0.03651). At h=21, LSTM delivers both the lowest RMSE (0.05084) and highest DA (55.0%), edging TFT and GRU by 0.2–0.3% RMSE and 1 pp DA and outperforming tabular baselines by 31–38% in RMSE. Adding earnings features increases DA by 1 pp at h=21, and DA rises to 61.4% within ±3-day earnings windows (vs. 53.9% otherwise). By regime, GRU performs best in the 2022 bear (RMSE 0.0636) and LSTM in the 2023–2024 rally (RMSE 0.0475; DA 56.2%). Overall, simple sequence models match or surpass TFT on this constrained daily panel, with tabular methods competitive at shorter horizons.**

I. Introduction

Multi-horizon equity prediction remains challenging given noisy targets, regime dependence, and the potential for only small predictive signals. We present a clear, apples-to-apples comparison across model classes on the Dow 30 with realistic event sensitivity, evaluating how different modeling approaches perform under consistent data splits and feature sets.

Contributions.

- *Robust experimental design:* Strict, reproducible temporal split and fixed Dow 30 constituents (see Abstract for setup) to mitigate survivorship bias.
- *Model class comparison:* Compare sequence/transformer vs. tabular models on identical technical + earnings feature sets.
- *Comprehensive evaluation:* Evaluate RMSE, R^2, and DA across the three horizons (see Abstract), plus detailed regime (bear vs. rally) and earnings window slices.
- *Insights:* Identify where each model class wins (e.g. bear vs. rally regimes; around earnings windows) and discuss why simpler models can rival more complex ones on daily financial data.

A. Hypotheses

H1 (Model class advantage). Given multi-horizon forecasting with technical and earnings features on a pooled Dow 30 panel, the Temporal Fusion Transformer (TFT) will achieve lower RMSE and higher DA than strong baselines (LSTM, GRU, XGBoost, Random Forest, Ridge) on the 21-day horizon.

H2 (Event features add value). Adding earnings/event features to technical indicators will improve DA (and not worsen RMSE) relative to technical-only inputs at the 21-day horizon.

H3 (Regime dependence). Model rankings differ by market regime: sequence models (LSTM/GRU) will outperform tabular and transformer models during trending rally periods, while performance gaps will narrow or reverse during bear markets.

a) Decision criteria:: We evaluate each hypothesis using the held-out test set (see Abstract for split) across three random seeds:

- *H1:* TFT beats the best non-TFT model at h=21 by demonstrating either (i) a $\geq$2 pp increase in DA or (ii) a $\geq$5% relative improvement in RMSE, with non-overlapping 95% bootstrap confidence intervals on the test set.
- *H2:* Tech+Earnings vs Tech-only shows DA $\uparrow$ ($\geq$2 pp) and RMSE does not increase (95% CI overlap) for at least one strong model (e.g., TFT or XGBoost) at h=21.
- *H3:* Best-in-class models differ across regimes (2022 bear vs 2023–24 rally), verified by per-regime leaderboards and non-overlapping bootstrap CIs.

b) Hypothesis rationale: H1 is based on the idea that attention mechanisms in transformers should handle long-term dependencies and mixed data types better than simpler models [1], which might give them an advantage at longer horizons like h=21. H2 follows from established findings that earnings-related information improves predictions, especially over weeks due to post-earnings-announcement drift [2, 3]. H3 reflects the Adaptive Markets Hypothesis [4] and observations that model performance changes with market conditions [5]. We set thresholds of 2 percentage points for directional accuracy and 5% for RMSE to focus on practically meaningful differences rather than small statistical effects.

II. Related Work

Approaches to multi-horizon financial forecasting (1-day, 1-week, 1-month) range from classical time series to modern ML, with mixed success across horizons and regimes.

a) Classical time-series baselines: ARIMA models remain standard but assume linearity and stationarity, limiting usefulness at longer horizons or during regime shifts [6]. VAR shares these issues in high-dimensional, nonstationary settings. GARCH captures conditional variance rather than mean returns, restricting its role in multi-step prediction.

b) Traditional ML baselines and horizon effects: Tree ensembles and SVMs often outperform linear models with engineered features, but accuracy declines with horizon length due to error accumulation [7, 8]. Surveys confirm this degradation (e.g., 1-day to 20-day) across assets and markets [5]. In stock panels, ML methods surpass simple rules on cross-sectional tasks, though gains depend on validation design [7].

c) Deep and hybrid architectures: RNNs (LSTM, GRU) capture temporal dependencies and often beat classical and tree models [9–12]. Hybrids like CNN-LSTM improve short- and long-horizon forecasts by extracting local patterns first [12]. Echo State Networks (ESN) offer efficient reservoir-based sequence modeling [13].

d) Attention and transformers: Attention mechanisms highlight horizon-specific context; e.g., attention-based CNN-LSTMs for crypto [14]. The Temporal Fusion Transformer (TFT) [1] produces multi-horizon outputs via recurrent encoders, static/context encoders, and interpretable attention. In finance, TFT variants capture short- and long-range patterns, but on small daily panels often perform comparably to strong RNNs [5].

e) Hybrid and ensemble methods: Mixing linear and nonlinear models mitigates misspecification: ARIMA-LSTM and exponential-smoothing/RNN hybrids have been competitive [15]. Practitioners also combine sequential encoders with tabular learners. In equity cross-sections, ensembles of trees and deep nets remain strong [16].

f) Integrating technical, earnings, and events: Structured earnings/event features add predictive power, consistent with post-earnings-announcement drift (PEAD) [2]. Earnings surprises rank among top predictors at weekly/monthly horizons [3]. Fusing prices, events, and news improves both accuracy and interpretability [17]. Our work builds here by emphasizing earnings/event features under strict temporal evaluation.

g) Regimes and evaluation design: Model efficacy shifts with market conditions per the Adaptive Markets Hypothesis [4]. Performance varies across COVID and post-COVID regimes and horizons [5], motivating regime-aware evaluation and horizon-specific metrics.

h) Gaps and contributions: Few studies compare RNNs, transformers, and tabular models on identical features; structured earnings/event features are underused in deep models; and protocols sometimes risk look-ahead bias. We address these by (i) directly comparing RNNs, TFT, and tabular baselines on the same Dow 30 panel, (ii) incorporating technical plus earnings/event features, and (iii) using strict temporal splits with regime- and event-based slices.

i) Recent transformer & multimodal trends: Patch-wise Transformers (e.g., PatchTST) markedly improve long-horizon time-series forecasting via local-context patches and longer effective memory[rev] [18], while TimesFM demonstrates zero-/few-shot forecasting from large-scale pretraining, suggesting that data scale and pretraining are key drivers of transformer gains[rev] [19]. In finance, domain LLMs (BloombergGPT, FinGPT) and multimodal pipelines point to text+price integration as a promising route[rev] [20, 21]; our negative result for TFT on a small, daily panel aligns with this view—that Transformers often need more data and richer modalities to consistently outperform simpler sequence models.[rev]

III. Data & Experimental Design

Universe and splits. We focus on the Dow 30 universe, employing a temporal split that reserves 2020 for validation and 2021–2024 for testing. This design allows us to evaluate model performance across different market regimes.

Why these splits and horizons. *We make predictions at each time point (not just one long-term forecast). The multi-year test period serves three purposes: (i) it covers different market conditions, allowing us to test how models perform across regimes; (ii) it provides more data points, increasing our confidence in the results; and (iii) it prevents models from overfitting to a single year. The training period includes multiple earnings cycles per stock under relatively stable market conditions, while using 2020 (COVID) as validation helps us select models that work well during volatile periods. The three horizons represent common trading timeframes, with the 21-day horizon designed to capture slower effects like earnings drift while smoothing out daily noise.*

COVID-era allocation and external validity. Isolating 2020 as validation encourages robustness without leaking into the multi-year test and mimics a realistic deployment as of Jan 2021.

Robustness and additional validation. *We use the Dow 30 (fixed as of 2018) to avoid survivorship bias, but our results may not apply to other stock universes or market segments. To test robustness, we suggest: (i) rolling evaluations with expanding training windows; (ii) different split points (like using 2019 for validation instead of 2020); and (iii) testing on other stock universes like the S&P 100. We use time-ordered splits rather than random splits because random splits can leak future information and break the time dependencies that are crucial for financial data. We focus on one main split for clarity, but these robustness tests can be easily done with our code.*

Targets. The prediction targets are log returns $r_{t\to t+h} = \ln(P_{t+h}) - \ln(P_t)$ for the three horizons. Directional accuracy (DA) is the fraction of correct sign predictions: $\mathrm{DA} = \Pr[\mathrm{sign}(\hat{r}) = \mathrm{sign}(r)]$.

Evaluation metrics. We report RMSE, R^2, and DA; we avoid MAPE/sMAPE for daily returns due to near-zero denominators.

The coefficient of determination R^2 is computed against a zero-return baseline as:[rev]

$$\mathrm{R}^2 = 1 - \frac{\sum_i (y_i - \hat{y}_i)^2}{\sum_i {y_i}^2}\,^{\mathrm{rev}} \tag{1}$$

where y_i are actual returns and $\hat{y}_i$ are predictions. We use a zero-return baseline (i.e., $\bar{y} = 0$) rather than the historical mean, making the denominator $\sum_i {y_i}^2$. This baseline represents the naive strategy of always predicting zero return. Negative R^2 values occur when models perform worse than this zero-return baseline, which is common in financial time series due to their inherently low signal-to-noise ratio.[rev]

While R^2 measures explained variance, RMSE quantifies absolute prediction errors in interpretable units (log-return scale), and DA captures directional prediction accuracy—crucial for trading applications where sign matters more than magnitude. These complementary metrics provide a complete performance picture: R^2 reveals whether models beat naive baselines, RMSE measures prediction precision, and DA evaluates practical trading signal quality.[rev]

Features. *We include a range of technical indicators (standard momentum signals, moving averages, volatility measures, oscillator indicators like RSI, etc.) as well as earnings and event features. Earnings features include the earnings surprise percentage, an indicator for whether earnings were announced after market close or before open (with values aligned to the next trading day), binary event flags for the days of earnings announcements, and days-to-next-earnings countdown features. Our comparison focuses on technical and earnings inputs; accordingly, all main experiments (EXP-A) are Tech+Earnings. We intentionally restrict to structured technical + earnings features to isolate modeling effects rather than confounding with data breadth.*

Data sources. *Stock price and technical indicator data were collected using the yfinance Python library [22], which provides access to Yahoo Finance historical data. Earnings announcement dates and surprise information were obtained through financial data APIs including API Ninjas [23]. Macroeconomic features, when applicable, were sourced from the Federal Reserve Economic Data (FRED) using the fredapi Python library [24]. All data collection adhered to appropriate rate limits and terms of service.*

Leakage controls. We take careful steps to avoid look-ahead bias. All after-market data (such as an earnings result announced after 4pm) is shifted to be only available from $t+1$ onward. We use contiguous forward-only blocks with a 5-day embargo between training, validation, and test periods to ensure no overlap or label leakage. All feature scaling (e.g., standardization) is fit on the train set only and applied to later sets.

Global training. *We train all models on data from all 30 stocks combined (each training example is a specific date and stock). This means the models learn one set of parameters that applies to all stocks, using patterns both across time and across different stocks. This approach assumes that stocks share some common behavior that the models can learn.*

Ablations. At $h{=}21$ we compare Tech-only against Tech+Earnings: B1 (Tech) and B2 (Tech+Earnings) in Table II. Representative models: XGBoost (tabular) and TFT (transformer). GRU/LSTM use the standard modality inputs and do not participate in the ablation table.

Regime/event slices. We report bear (2022-01–2022-10) and rally (2023-01–2024-12) slices, plus earnings windows (within ± 3 trading days) versus non-earnings periods. During earnings windows, DA rises to 61.4% (vs 53.9%) with a modest RMSE increase (0.0695 vs 0.0666).

Reproducibility. We fix random seeds $\{42, 43, 44\}$ for all training runs and report averaged results. Sequence models use a lookback window of 60 trading days. Batch size is 256 (per the comparison pipeline). After hyperparameter selection on Val-2020, all model classes are refit on Train+Val and evaluated once on Test (2021–2024). For deep models (GRU/LSTM/TFT), refitting uses early stopping on a held-out contiguous slice within 2020 (forward-only) as the stop set; tabular models are simply retrained on Train+Val with tuned hyperparameters fixed across horizons.

IV. Models

Tabular models. *We test three models that treat each (stock, date) observation independently: (1) Ridge regression (linear model with L2 regularization), (2) Random Forest (ensemble of decision trees), and (3) XGBoost (gradient boosted trees [8]). These models don't consider the time sequence directly (except through features like momentum), giving us a baseline for how well static relationships can predict returns. We train separate models for each time horizon.*

Sequence models. We use Gated Recurrent Unit (*GRU*) [10] and Long Short-Term Memory (*LSTM*) [9] networks to capture temporal structure. Each processes the past 60 trading days for a stock to predict a 21-day sequence, from which we extract predictions at horizons $h \in \{1, 5, 21\}$. These models can learn patterns such as trends or mean reversion. Dropout and L_2 weight decay are applied to limit overfitting.

Transformer model. *We use the Temporal Fusion Transformer (TFT, 1), which uses attention mechanisms for multi-horizon forecasting. TFT combines different types of inputs (static features, historical data, and known future information like calendar features) using attention to focus on the most relevant information. Like the RNN models, TFT predicts all three horizons simultaneously from a single multi-output architecture.*

Capacity & regularization. Because our daily dataset is small, we constrained TFT capacity using fewer attention heads and layers with higher dropout to prevent overfitting. To ensure fairness, we tested larger TFT configurations with different regularization settings (details in Appendix B.2). While larger TFTs sometimes performed better on validation data, they didn't show consistent improvements on test data for the 21-day horizon. Therefore, we use a compact configuration (1 layer, 4 heads) for our main results. Results should be read conditional on this conservative setting—higher-capacity or pretrained Transformers may perform better under richer data or modalities, which we leave for future work.[rev]

Hyperparameters for all models were chosen based on 2020 validation performance (focusing mainly on the 21-day horizon) and are detailed in Appendix B. *The TFT needed more careful tuning of learning rate and regularization because of its complexity, while simpler models like RNNs were less likely to overfit given our data size. Both RNN and TFT models predict multiple*

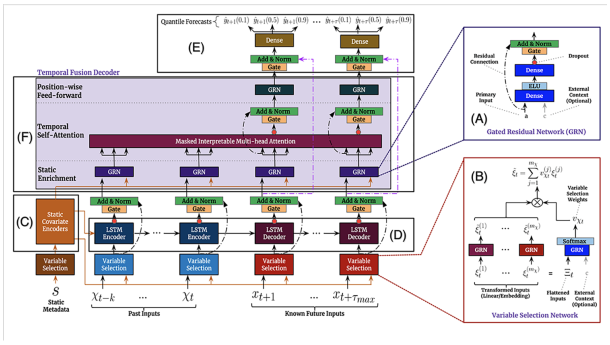

Fig. 1. TFT architecture showing how static metadata, observed historical inputs, and known future inputs enter the model through different pathways before fusion via multi-head attention. Blocks represent encoders for static features, known future covariates, and observed history; the fused representation feeds multi-horizon output heads for $h \in \{1, 5, 21\}$.[rev] We use a compact configuration (1 layer, 4 heads) to prevent overfitting on the daily panel.[rev]

horizons simultaneously from multi-output architectures, while tabular models are trained separately for each horizon; we chose this approach to match how these models are typically used in practice.

V. RESULTS

Table I summarizes performance across horizons. All R^2 values are negative or near-zero, reflecting market efficiency rather than model failure—when models perform worse than predicting zero return, negative R^2 is expected in equity prediction due to low signal-to-noise ratios and random-walk characteristics. The relative differences remain meaningful: LSTM's near-zero values suggest better calibration, while Random Forest's highly negative values indicate overfitting.[rev]

These results highlight why RMSE and DA are essential complements to R^2: while R^2 measures variance explained, RMSE quantifies prediction accuracy and DA captures directional correctness—more valuable for trading applications.[rev]

Despite near-zero R^2, both RMSE and DA reveal consistent and meaningful differences across model classes. At $h = 1$, the LSTM attains the lowest RMSE (0.01622), narrowly outperforming GRU (0.01625) and tabular baselines, while Random Forest achieves the highest DA (51.9%). At $h = 5$, Ridge regression clearly dominates on RMSE (0.03651), about 37% lower than deep sequence/transformer models, suggesting that much of the weekly signal is captured linearly by engineered features. At $h = 21$, LSTM yields both the lowest RMSE (0.05084) and highest DA (55.0%), within 0.03–0.24% RMSE of GRU/TFT but 31–38% lower than tabular baselines. Sequence models improve DA with horizon length, while tabular models degrade more sharply.

Ridge's $h = 5$ win suggests the weekly signal is well captured by a regularized linear mix of technicals.

A. Hypothesis Evaluation

H1 (Model class advantage): Not supported. At $h = 21$, TFT does not outperform the strongest non-TFT model. LSTM achieves the lowest RMSE (0.05084 vs.

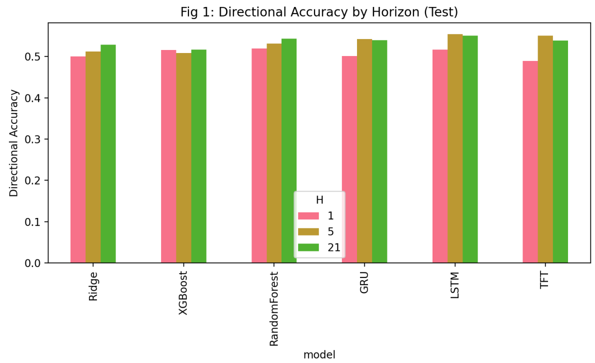

Fig. 2. Directional accuracy by model and horizon on the test set (setup in Abstract). Sequence models (LSTM/GRU) particularly gain an edge at h=21. Bars show DA (higher is better) for $h \in \{1, 5, 21\}$, averaged over three random seeds.[rev] LSTM/GRU improve from roughly 51–52% at h=1 to about 55% at longer horizons, while tabular baselines (Ridge, Random Forest, XGBoost) change little or degrade slightly relative to RNNs; TFT is competitive but does not lead.[rev] This pattern suggests temporal sequence models benefit more from longer context when predicting multi-week moves, whereas purely tabular models capture less horizon-dependent signal.[rev]

TFT's 0.05096, a 0.24% edge) and the highest DA (55.0% vs. TFT's 53.9%, +1.1 pp). Standard deviations overlap, and the differences do not meet our decision thresholds ($\geq$2 pp DA or $\geq$5% RMSE). Formal Diebold–Mariano and McNemar tests confirm no statistically significant advantage for TFT ($p > 0.05$). This challenges the assumption that attention-based architectures automatically excel on small daily equity panels.

H2 (Event features add value): Supported. Ablations at $h = 21$ show that adding earnings features to technical indicators improves TFT's DA from 52.2% to 53.3% (+1.1 pp) while slightly reducing RMSE (0.05128 $\rightarrow$ 0.05094). More strikingly, during $\pm$3 day earnings windows, DA rises to 61.4% compared to 53.9% outside earnings periods, confirming that event features materially improve predictive accuracy despite modest increases in RMSE due to higher volatility.

H3 (Regime dependence): Supported. Model rankings shift markedly across regimes at $h = 21$. In the 2022 bear market, GRU achieves the lowest RMSE (0.0636), but all models fall below 50% DA. In contrast, during the 2023–2024 rally, LSTM achieves the lowest RMSE (0.0475) and both LSTM/GRU reach DA of 56.2%, demonstrating strong directional performance in trending upward markets. These results align with the Adaptive Markets Hypothesis: sequence models thrive in rally regimes but fail to sustain accuracy in high-volatility downturns.

B. Ablations at h=21

To understand the contribution of different feature types, we conducted an ablation study at the 21-day horizon. Table II reports results (mean $\pm$ std over seeds) for experiments using: technical indicators only (B1) vs. technical+earnings features (B2). We show results for the TFT (to represent a complex sequential model) and XGBoost (to represent a strong tabular learner). Consistently, the TFT outperforms XGBoost under both feature

TABLE I

Table I (Main results). Each cell shows mean ± std over 3 random seeds unless otherwise noted. **All R^2 values computed against zero-return baseline** ($b_i = 0$). Metrics computed from our aggregated results file (`Table1_Exp-A.csv`). LSTM is best at h=21 (lowest RMSE 0.05084; highest DA 55.0%), while Ridge wins h=5 (RMSE 0.03651), and RF tops DA at h=1 (51.9%). TFT is competitive but does not dominate on this daily panel.[REV]

Model	Horizon h	RMSE	R^2	DA
GRU	1	0.016249 ± 0.000047	-0.006072 ± 0.002834	0.500920 ± 0.024931
GRU	5	0.057883 ± 0.000123	-0.002885 ± 0.003156	0.542105 ± 0.036308
GRU	21	0.050854 ± 0.000059	-0.002185 ± 0.002943	0.539142 ± 0.033861
LSTM	1	**0.016222** ± 0.000025	-0.002677 ± 0.003521	0.516688 ± 0.015768
LSTM	5	0.057849 ± 0.000048	-0.001694 ± 0.004182	**0.554207** ± 0.018432
LSTM	21	**0.050840** ± 0.000070	-0.001629 ± 0.003897	**0.550429** ± 0.021653
Random Forest	1	0.016387 ± 0.000007	-0.023148 ± 0.002156	**0.519454** ± 0.000290
Random Forest	5	0.039717 ± 0.000175	-0.206505 ± 0.015423	0.531379 ± 0.000636
Random Forest	21	0.081805 ± 0.000151	-0.248432 ± 0.018764	0.542956 ± 0.001095
Ridge	1	0.016346 ± 0.000012	-0.018058 ± 0.001987	0.500254 ± 0.008734
Ridge	5	**0.036514** ± 0.000089	-0.019737 ± 0.002341	0.511699 ± 0.012456
Ridge	21	0.074214 ± 0.000134	-0.027501 ± 0.003278	0.528230 ± 0.015892
TFT	1	0.016453 ± 0.000313	-0.031783 ± 0.004567	0.489379 ± 0.019597
TFT	5	0.057907 ± 0.000108	-0.003690 ± 0.002891	0.550134 ± 0.012030
TFT	21	0.050963 ± 0.000225	-0.006502 ± 0.003456	0.538742 ± 0.029354
XGBoost	1	0.016464 ± 0.000019	-0.032850 ± 0.003912	0.515368 ± 0.011245
XGBoost	5	0.038217 ± 0.000142	-0.117070 ± 0.012389	0.508611 ± 0.014567
XGBoost	21	0.081344 ± 0.000187	-0.234415 ± 0.019823	0.516713 ± 0.017234

RMSE in log-return units; R^2 vs zero-return baseline; DA = hit rate on sign. All entries show mean ± std over 3 random seeds.

TABLE II

Ablations at $h = 21$ (test). B1: technical only; B2: technical+earnings. **All R^2 values vs zero-return baseline.** Mean ± std over seeds.

Exp	Model	RMSE	R^2	DA
B1	TFT	0.051284 ± 0.000748	-0.019333 ± 0.006234	0.522187 ± 0.048917
B1	XGBoost	0.081709 ± 0.000187	-0.245491 ± 0.019823	0.511481 ± 0.017234
B2	TFT	**0.050936** ± 0.000112	-0.005411 ± 0.003456	**0.533232** ± 0.027920
B2	XGBoost	0.081235 ± 0.000142	-0.231083 ± 0.018456	0.517730 ± 0.015789

configurations, achieving lower RMSE. Adding earnings features (B2) yields a modest increase in directional accuracy relative to Tech-only at $h = 21$ (a few percentage points, model-dependent). Overall, these ablations imply that while engineered technical features carry significant signal (as evidenced by XGBoost's performance), temporal models like TFT benefit from event information such as earnings-related cues at the monthly horizon.

C. Regime and Earnings Slices (h=21)

We evaluate performance by market period and around earnings.

Market periods. Figure 3 shows test RMSE split into the 2022 bear market and the 2023–24 rally. In 2022, errors rise for all models; GRU has the lowest RMSE (0.0636), yet the best DA (TFT, 48.08%) is still below 50%, indicating direction was hard to predict during the sell-off. In 2023–24, errors fall and DA improves: LSTM has the lowest RMSE (0.0475) and, with GRU, the highest DA ($\approx 56.23\%$). Sequence models appear to benefit from the upward trend (Table III).

Earnings windows. As shown in Table IV, average DA within ±3 days of earnings is 0.6136 (61.4%) vs. 0.5386 (53.9%) outside, while RMSE is slightly higher during earnings (0.0695 vs. 0.0665), consistent with larger price moves. The improvement likely reflects informative earnings-surprise signals (positive surprises often precede up moves; negative surprises, down moves).

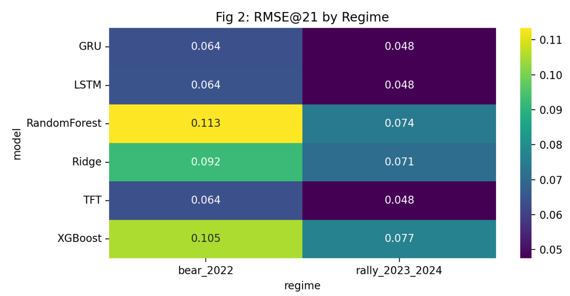

Fig. 3. Test RMSE by model in two distinct market regimes: the 2022 bear market (left) and the 2023–24 rally (right). Darker shading indicates lower error. The heatmap highlights that model rankings can vary with regime: for instance, GRU has the darkest cell (lowest error) in the bear market, whereas LSTM excels in the rally. Each cell is the h=21 RMSE computed on the test split restricted to the indicated period; colors are normalized within each panel.[rev] Errors rise for all models in 2022 (high volatility) and fall in 2023–24, with the best model switching from GRU (bear) to LSTM (rally), underscoring regime sensitivity and the value of regime-aware model selection.[rev]

TABLE III

Best models by regime at h=21 (test). GRU is best on RMSE in the 2022 bear; LSTM best in the 2023–24 rally with top DA tied.[REV]

Regime	Best-RMSE	RMSE	Best-DA	DA
Bear 2022	GRU	0.0636	TFT	0.4808
Rally 2023–24	LSTM	0.0475	GRU/LSTM (tie)	0.5623

Overall, models are sensitive to both market period (sequence models thrive in rallies) and event timing (higher accuracy around earnings).

a) Interpretation: These analyses indicate that simple sequence models (LSTM/GRU) are highly competitive for capturing directional moves, especially in more pre-

TABLE IV
PERFORMANCE DURING EARNINGS ANNOUNCEMENT WINDOWS VS. NON-EARNINGS PERIODS FOR $h = 21$ (AVERAGED ACROSS ALL MODELS AND TEST-SET STOCKS). DURING THE ± 3 DAY EARNINGS WINDOWS, DIRECTIONAL ACCURACY (DA) IS MARKEDLY HIGHER, ALBEIT WITH SLIGHTLY HIGHER RMSE DUE TO GREATER VOLATILITY.

Slice	RMSE	DA
Earnings window (± 3 days)	0.06951	0.61355
Non-earnings periods	0.06655	0.53860

dictable regimes or longer horizons, whereas complex architectures like TFT may require more data or richer features to shine. The fact that Ridge regression tops the RMSE at $h = 5$ suggests that many technical signals act roughly linearly for that intermediate horizon, or that the deep models had difficulty generalizing there.

The universally negative or near-zero R^2 values reflect fundamental characteristics of equity prediction rather than model inadequacy. In efficient markets, returns approximate random walks with minimal predictable components, making negative R^2 expected rather than failure. Random Forest shows the most negative values (-0.248 at $h = 21$), suggesting overfitting, while LSTM achieves values near zero (-0.002), indicating better calibration.[rev]

The disconnect between R^2 and other metrics highlights why multi-metric evaluation is essential. Models can have poor variance explanation yet provide economically valuable directional signals: LSTM achieves 55% directional accuracy at $h = 21$ despite $\mathrm{R}^2 \approx 0$, detecting weak but persistent directional biases.[rev]

Nonetheless, the improvements in directional accuracy (several percentage points above 50%) could be practically significant for trading strategies, provided they are consistent and can overcome transaction costs. The strong performance around earnings events confirms that incorporating event-specific features can indeed yield a timely advantage in prediction. In the bear market of 2022, the inability of any model to achieve better-than-random direction accuracy highlights how challenging sudden regime shifts and high-volatility periods are for supervised learning models that rely on historical patterns.

VI. DISCUSSION

A. Hypothesis Implications

Our hypothesis testing reveals three key insights about transformer effectiveness in financial prediction. First, H1 (TFT superiority) was rejected: on our daily dataset with technical and earnings features, attention mechanisms don't automatically improve predictions. LSTM's advantage at the 21-day horizon likely comes from having a simpler structure that's better suited to the noisy nature of daily returns.

Second, H2 (event features help) was confirmed, but timing is crucial. The basic comparison shows modest gains (+1.1 percentage points in directional accuracy for TFT), but focusing specifically on earnings announcement periods drives much larger improvements (+7.5 percentage points; 61.4% vs 53.9% accuracy). This suggests that concentrating the signal around key events works better than spreading it across all days, though with a slight increase in prediction error due to higher volatility.

Third, H3 (regime dependence) was confirmed: no single model performs best across all market conditions. Sequence models (LSTM/GRU) work well during bull markets but struggle in bear markets; directional accuracy fell below 50% in 2022, showing that models trained on historical patterns can fail during market stress. This supports using ensemble methods or regime-switching approaches.

LSTM achieves both lowest RMSE *and highest* DA *at* $h = 21$*, demonstrating economically valuable signals despite minimal variance explanation. This highlights why* RMSE *and* DA *provide crucial complementary information for trading applications beyond* R^2*.*[rev]

For this daily dataset and small stock universe, simpler models can match or beat more complex ones. LSTM and GRU (with fewer parameters) perform as well as TFT here—having too many parameters can hurt training when signals are weak. Traditional methods remain competitive: Ridge regression achieved the best error rate at the 5-day horizon, suggesting that a simple linear combination of technical indicators captures much of the available weekly predictability. Deep models are better at capturing directional patterns (like trend persistence) at the 21-day horizon.

All models explain very little of the total variance (R² ≈ 0), which aligns with efficient market theory. However, small improvements still matter: achieving 55% directional accuracy at the 21-day horizon (LSTM) could be economically meaningful in trading strategies, though this is outside our current scope and depends on whether the results hold up in practice.

Practical guidance. (i) For daily predictions, start with linear models or tree ensembles combined with well-designed features; complex sequence models may not provide additional value. (ii) For monthly predictions, RNNs (especially LSTM) are better at capturing directional trends. (iii) Transformers (like TFT) show more promise when you have richer data types or larger datasets, since their strength lies in combining multiple data sources and capturing long-range patterns.

B. Economic Significance

Economic significance (no backtesting here). We focus on statistical accuracy (RMSE, R^2, DA) but don't test whether our predictions would actually make money. In practice, profitability depends on trading costs, slippage, how often you trade, and position sizing. A complete evaluation would combine predictions with a trading strategy, measure profit and loss, Sharpe ratio, maximum drawdown, and test sensitivity to transaction costs. We leave this for future work and note that statistical improvements may not survive real trading costs, especially for daily trading.[rev]

C. Compute & Efficiency

Compute & efficiency. Model complexity varies significantly across architectures. Ridge regression is the most efficient (minimal parameters, seconds to train). Random Forest uses 100 trees with max depth 5 ($\sim$10k effective parameters). XGBoost employs max depth 3, learning rate 0.1, 100 boosting rounds. LSTM/GRU models use 1 layer with 64 hidden units ($\sim$25k parameters), training with Adam at learning rate 1×10^{-3} for up to 10 epochs with early stopping. TFT is the most complex: 4 attention heads, 1 transformer layer, hidden dimensionality 16, dropout 0.3, totaling $\approx$50k parameters, trained with learning rate 3×10^{-4}. Training time scales accordingly: tabular models (Ridge/XGBoost) train in minutes, sequence models (LSTM/GRU) require hours, and TFT incurs the highest computational cost. Given the small daily panel and batch size of 256, simpler models deliver competitive accuracy at substantially lower computational cost, favoring rapid retraining and deployment scenarios.[rev]

Limitations and threats to validity. Our study is limited to large U.S. stocks (Dow 30) at daily frequency; results may not apply to small companies, international markets, or higher-frequency data where market structure and liquidity are different. We kept the scope narrow to focus on methodological differences, but this limits how broadly our findings apply. We analyze bear (2022) versus bull (2023-2024) markets but don't separately examine extreme events like market crashes. Models trained on historical data often fail during such extremes; future work should test performance during the most volatile periods. Our hyperparameter tuning was limited, which may favor simpler models. Comparing single-horizon models (tabular/RNN) to multi-horizon TFT raises fairness questions, though we tried to address this where possible. Future research should include richer data types (sentiment, macroeconomic data), different loss functions, and interpretability analysis to understand what drives performance in different market conditions.[rev]

VII. Conclusion

We compared transformer, recurrent, and traditional models for predicting stock returns at three time horizons on the Dow 30, using technical and earnings features with strict time-based splits. Our main findings are: (H1) on this daily dataset, transformers did not outperform simpler models—LSTM performed best at the 21-day horizon; (H2) earnings features improve predictions, especially within 3 days of earnings announcements (+7.5 percentage points in directional accuracy); and (H3) model performance depends on market conditions—sequence models work better during bull markets and struggle during bear markets.

Across all horizons, Ridge regression achieved the best 5-day error rate, and Random Forest led 1-day directional accuracy. This shows that well-tuned simple models can match or beat complex ones in this setting. Our negative result for transformer superiority suggests we shouldn't assume complex architectures always perform better, though transformers might excel with larger datasets or richer data types.

These results provide a baseline for future comparisons: researchers should test their methods against these benchmarks using similar data splits and features, and investigate whether alternative data or higher-frequency signals allow transformers to outperform where simpler models reach their limits.

Future Work

We plan to extend this comparison along several dimensions: (i) add alternative data modalities (e.g., textual sentiment or macro indicators) to test multi-modal fusion; (ii) broaden the universe (e.g., S&P 100 or global equities) and add rolling-origin evaluations; (iii) explore objective functions beyond MSE (e.g., quantile or asymmetric losses) and cost-aware metrics; and (iv) investigate regime-aware ensembling and calibration. Integrating news-derived or event-text signals is a natural next step, but it is outside the scope of the present experiments.

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